Department of Computer Science and Engineering
Academic Year 2021 – 2022(Odd / ~~Even~~ Semester)
Degree, Semester & Branch: B.E, I & CIVIL
Course Code & Title: GE3151 & Problem Solving and Python Programming
Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic: Unit I /** Illustrative Problems (Guess an Interger)
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 1,2,3
- **Activity Chosen: Role Play**
- **Justification:**

  A number guessing game is a simple guessing game where a user is supposed to guess a number between 0 and N in a maximum of 10 attempts. The game will end after 10 attempts and if the player failed to guess the number, and then he loses the game. This active learning strategy is chosen to help students to understand the concept of implementing an algorithm for guessing an integer problem.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

  The course instructor chosen two students namely student 1 (Mr.Sanjaykumar) and student 2 (Mr.Esaki Hariganesh) and also he assigned the role for student 1 as no. Generator who generates the random no between 1 to 100 and student 2 as guesser who guesses the no between 1 to 100 as shown in fig.1. Student 2 guesses the no and inform the no to student1as shown in fig.2. Student 1 checks the no he received from student 2 with the no. Generated by him and say whether no is matched or greater or less as shown in fig.3. Bsased on the rely from student 1, student 2 guess another no and inform to student 1. This process is repeated until 10 times. With in 10 time student 2 guess the no correctly, he is a winner or otherwise he is loser. After completion of this roleplay, again the course instructor explained the concpets.

Fig.1. Mr. Sanjay kumar (Student 1) generates the random no between 1 to 100.

Fig.2. Mr.Esaki Hariganesh (Student 2) guessed the number and reply to Student 1

Fig.3. Student 1 cehcks the no with already generated no.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO 10	PO12	PSO 3
CO 1	3	2	2	1	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 1**	**PO 2**	**PO 3**	**PO 5**	**PO 12**
	3	2	1	**1**	**2**
Justification for correlation	Students could be able to apply basic engineering theory and principles to solve the guess an integer problem	Students could be identify their own example to explain the algorithm	Students could be able to recollect the basic principles of guess an integer no algorithm	Students could able choose the modern tool to interpret and explain the algorithm for guess an integer number	Students interacted and shared their views about the concept with their peer group by ethically

Innovative practice	**PSO 3**
	2
Justification for correlation	Students could be able to provide the solution for real time problems.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students felt that this activity is very useful and got good idea about the development of algorithm for guessing an integer number
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - Most of the stuents understand and write the step by step procedure of guessing an integer number individually.
    - Students interaction with their peer members and the class during the activity was good and effective.
  - ***Challenges faced in implementation:***
    - Due to time limitation, only few pairs of students were made to do the activity.

References:

- https://www.kent.edu/ctl/think-pair-share
- Johnson, D., Johnson, R. (1999). Making Cooperative Learning Work: Theory into Practice, 38 (2), 67-73. Fitzgerald, D. (2013). Employing think-pair-share in associate degree nursing curriculum. Teaching & Learning in Nursing. 8, 3, p 88-90.
- Goodwin, M. (1999). Cooperative learning and social skills: What skills to teach and how to teach them. Intervention in School & Clinic, 35 (1), 29.
- Lyman, F. (1981). The responsive classroom discussions: the inclusion of all students. A. Anderson (Ed.), Mainstreaming Digest, College Park: University of Maryland Press, pp. 109-113.
- Mr.R.Venkatesh, "Introduction to Yoga and Meditation for professional excellence and stress management " in course GE 6075 Professional Ethics in Engineering, https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf

Signature of Faculty Member **HOD**